Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/08/2025
PI Name:	Sackett, Charles Ph.D.		
Project Title:	Development of Atom Interferometry Experiments for the International Space Station's Cold Atom Laboratory		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22904-1000	**Congressional District:**	5
Comments:	Other names: CA Sackett; Cass Sackett.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2025 per D. Griffin/NASA-HQ (Ed., 1/20/25). NOTE: Contract number changed to JPL 1700183 per J. Callas/JPL (Ed., 8/29/24). NOTE: End date changed to 9/27/2024 per U. Israelsson/JPL (Ed., 1/6/22) NOTE: Extended to 9/30/2022 per U. Israelsson/JPL (Ed., 3/9/21) NOTE: Extended to 10/28/2020 per PI (Ed., 2/28/2020) NOTE: Extended to 10/30/2019 per U. Israelsson/JPL (Ed., 12/14/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1700183 ; JPL 1502012		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The ultimate objective of this proposal is to develop an ultra-high sensitivity atom interferometer capable of operating in and benefiting from a microgravity environment. The interferometer would be specifically suited for measurements of rotations, but it would be broadly applicable to a variety of precision measurements. Ground and flight based efforts are proceeding in three broad areas. First, we are performing ground studies and developing a flight mission for the Cold Atom Laboratory (CAL) to study atomic techniques for inertial sensing in microgravity. Ground efforts include development of new rotation-sensing techniques and implementation of an optically suspended atom source for gravimetry. Flight efforts involve implementation and characterization of atom interferometry techniques using the CAL apparatus on the International Space Station (ISS). Second, we are investigating methods to produce an ultra-low temperature atom source in free space using the CAL apparatus. The apparatus produces atoms confined in a magnetic trap, but inertial measurements require free atoms. We will investigate releasing the atoms by gradually turning off the trapping fields, allowing the atoms to adiabatically expand and cool off. This can produce a relatively dense and very low-velocity sample that is ideal for atom interferometry methods. Third, we will continue ground-based studies to develop novel precision measurement techniques for use with atom interferometry, such as tune-out spectroscopy. Techniques like this are useful for advancing scientific knowledge and would be good candidates for future flight studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of precision inertial sensing techniques is useful for Earth-based as well as space-based navigation. Besides using direct sensing for inertial navigation, rotation sensing can also be useful for north-finding while gravity sensing can be used to tabulate local gravity variations and form a type of three-dimensional map for navigating. These techniques also have many applications in geophysics. Gravity sensing can be used for oil and mineral exploration, while rotation sensing can detect dynamics in the Earth's core. Gravity sensing also has defense applications such as locating underground tunnels and potential screening cargo for high-density contraband or weapons. Other precision measurement applications have less direct impact, but advance scientific knowledge. For instance, precision tune-out spectroscopy measurements of atomic matrix elements can be used to improve the interpretation of atomic parity violation experiments. These in turn impact our understanding of the standard model of particle physics and thus the nature of our universe. Direct benefits of such understanding can be hard to trace, but in general the continued advance of technological applications builds on advances in our fundamental knowledge.
Task Progress:	Flight efforts during the reporting period were constrained by apparatus problems, but thanks to the efforts of the NASA Jet Propulsion Laboratory (JPL) project team, some experiments were possible. Our efforts during this period focused on calibrating the magnetic field environment at the location of the atoms in the Cold Atom Laboratory (CAL) apparatus. The adiabatic expansion techniques that we have developed appear to be limited by a larger-than expected background field, but we have not, before now, attempted to measure this field directly. We are developing a technique for this based on rf spectroscopy, using the fact that the frequency interval between Zeeman states depends on the field. In order to find the three vector components of the background field, we apply in turn bias offsets in the known x,y, and z directions. For each direction, we apply first a positive and then a negative offset, and for each we measure the Zeeman transition frequency. By comparing the transition frequencies for the two cases, the background components in each direction can be determined. With a frequency resolution of about 1 kHz, we can determine the field with milliGauss resolution, which is adequate for our experiments. The same measurement campaign will provide an accurate calibration of the CAL bias field coils, which will also be useful. Although we did not have time to complete these measurements during the reporting period, we were able to demonstrate the technique. Unfortunately, new apparatus problems have arisen, but we are now well-positioned to carry out this calibration in the new SM-3X module, expected to launch in early 2026. Our ground-based work has centered on rebuilding our apparatus to enable new capabilities. Our previous system used in-vacuum magnet coils to produce our atom trap. This made the coils difficult to access for measurement or adjustment, and also distorted the AC fields they produced due to eddy currents in the surrounding vacuum chamber. We have now proceeded to the next phase of the apparatus development, in which the atoms will be transported into a small glass vacuum cell that will be surrounded by magnetic coils in air. The coil assembly will be easy to characterize and modify, and there are no nearby metallic surfaces to produce eddy currents. During the reporting period, we rebuilt the vacuum apparatus and added the new cell. We successfully baked the chamber and achieved suitable vacuum pressure. We reestablished the lasers and optics to produce a magneto-optical trap and load cold atoms into a magnetic trap. We achieve atom numbers comparable to our previous results. We have also constructed and characterized the magnetic assembly. Notably, the power handling capacity is sufficient to limit temperature increases to about 10 C at full current. Our next step is to transport atoms into the glass cell. This requires loading atoms into an optical tweezer laser trap, and then translating the trap into the cell.
Bibliography Type:	Description: (Last Updated: 04/22/2026)
Articles in Peer-reviewed Journals	Williams JR, Sackett CA, Ahlers H, Aveline DC, Boegel P, Botsi S, Charron E, Elliott ER, Gaaloul N, Giese E, Herr W. "Pathfinder experiments with atom interferometry in the Cold Atom Lab onboard the International Space Station." Nature Communications. 2024 Aug 13;15(1):6414. https://doi.org/10.1038/s41467-024-50585-6 , Aug-2024